

Modeling China's interprovincial electricity transmission under low carbon transition



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HIGHLIGHTS

- A 30-province electricity transmission model in multi-voltage levels is developed.
- Electricity transmission are explored under the NDC and 2-degrees targets.
- The low carbon transition will increase electricity transmission.
- The overall electricity transfer pattern will be much more complex in future.
- The current policies that promote ultra-high voltage lines need to be maintained.

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ABSTRACT

Interprovincial electricity transmission is expected to play an important role towards low carbon transition to facilitate new and renewable energy development in West China to meet the increasing electricity demand in East China. To analyze changes in interprovincial electricity transmission and transmission infrastructure construction under China's Nationally Determined Contributions and the 2-degree target, this study develops an interprovincial electricity transmission model with multi-voltage levels and integrates it with an improved 30-province energy system model to simulate reference scenario (REF) and three low carbon scenarios from a whole energy system perspective, CP30 (emissions peak in 2030), CPE (emissions peak in advance) and C2D (2-degree target). In 2050, electricity transmission is expected to increase to 2526.5 TWh, 3299.3 TWh, 3714.2 TWh and 4002.0 TWh under the REF, CP30, CPE and C2D scenarios, respectively, and the ultrahigh-voltage transmission demand is expected to reach 595.5 GW, 843.3 GW 917.0 GW and 1198.0 GW respectively. The overall pattern of electricity transmission will become increasingly complex as long-distance and large-capacity transmission becomes more prominent. Inner Mongolia, Northwest China and Southwest China will be the major sources of electricity transmissions, and the Beijing-Tianjin-Hebei region, Guangdong, Central China and East China will be the major importers. Ultrahigh-voltage lines will form large-capacity and long-distance electricity transmission paths connecting the power generation bases and load centers. These findings suggest that there will be a significant increase in the demand for electricity transmission and infrastructure construction in the future and that the policies promoting ultrahigh-voltage lines development should be enhanced.

1. Introduction

Under China's Nationally Determined Contributions (NDCs), the Chinese government prioritized reaching a national carbon emissions peak in 2030 or sooner, a carbon intensity decrease by 60–65% compared with the carbon intensity in 2005, and a renewable energy share of greater than 15% of primary energy use in 2020 and greater than 20% in 2030 [1]. The countries that signed the United Nations Framework Convention on Climate Change agreed to the goal of “holding

the increase in the global average temperature to well below 2 °C above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5 °C above pre-industrial levels” [2]. When implementing a response to climate change at the provincial level in China, the different contexts among provinces, such as the economic development level and energy reserves, should be considered. Fig. 1 shows the wind, solar, and hydropower reserves; population and GDP in 2015 for each province (Taiwan, Hong Kong and Marco are not shown due to statistical issues). Renewable energy substitution plays an important role in the low

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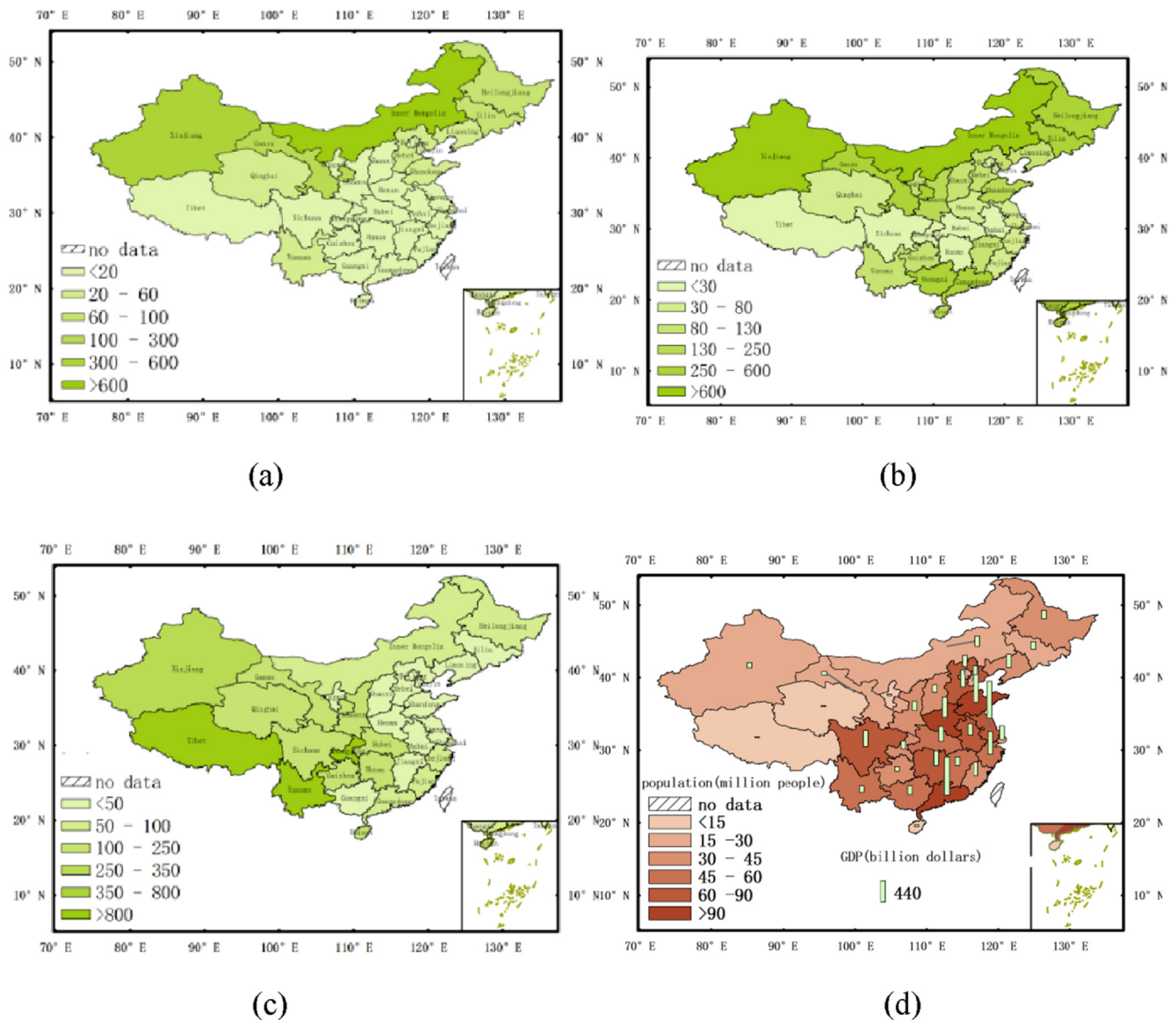


Fig. 1. (a) Potential wind capacity at 70 m (GW); (b) Potential solar capacity (GW); (c) Potential hydropower capacity (GW); (d) GDP and population in 2015.

carbon transition, but there are major differences in renewable energy reserves between provinces. Compared with eastern and southern China, western and northern China have considerably lower populations and larger renewable energy reserves. For example, in Inner Mongolia, the proportion of the national potential wind capacity at 70 m is 56.9%, and the proportion of the national potential solar capacity is 34.1%, with a population of 25.1 million in 2015. However, in Henan, which had a population of 94.8 million in 2015, the proportions of the potential wind and solar capacities are 0.15% and 0.29%, respectively [3]. Energy consumption is related to economic development and population [4]. GDP can be used in energy consumption projections [5,6], while electricity demand can be projected by historic economic statistics [7] or trend analysis methods [6]. These studies pointed out that electricity is related to GDP and population. It shows that there is distribution mismatch between renewable energy reserves and electricity demand.

With supported from low carbon development policies, there were significant increases in solar and wind power in Northwest China. However, wind and solar curtailment in 2015 reached 14.6% [8]. For example, the wind and solar curtailment in Gansu was 10.8 TWh, and the electricity exported from Gansu totaled only 24.4 TWh. The limited

interprovincial electricity transmission capacity is one of the reasons for significant wind and solar energy curtailment in the northern and western provinces [9]. To extend the interprovincial transmission capacity, the government invested in the construction of several high-capacity ultrahigh-voltage (UHV) lines. In 2017, with a new UHV line in operation, the total transmission capacity of 12 lines was 301 TWh [10]. In 2018, 20 UHV lines were put into operation, and the total transmission reached 398 TWh, of which 52.3% was for renewable energy transmission. However, as noted in the report by the Chinese Academy of Engineering, there were problems such as a low load rate, which resulted in a low transmission demand and limited economic benefits for some UHV lines [11]. For example, the designed capacity of the Zhejiang-Fujian line is 6.8 GW, but in 2018, the actual transmission of electricity on the Zhejiang-Fujian line reached only 6.9 TWh [12,13].

Since electricity demand, resource endowment and renewable energy substitution level associated with the low carbon transition vary across provinces and the demand for UHV electricity transmission infrastructure construction needs to be determined, it is worthwhile to analyze the electricity transmission situations and electricity transmission demands for UHV lines.

Electricity transmission plays an important role in the global low

carbon transition. In an international perspective, Joskow [14] pointed out that transmission capacity expansion is needed to decarbonize the electricity sector efficiently. Brancucci Martínez-Anido, Vandenberg [15] analyzed the cross-border electricity transmission in Europe and found that cross-border transmission capacity in Europe will reduce annual dispatch costs, and have limited impact on the security of supply. Boie, Kost [16] pointed out that substantial transmission grid reinforcements on alternating current and the construction of a high voltage direct current overlay grid are prerequisites for the high renewable energy deployment in North Africa and Europe. The results of Wang, Zhang [17] indicate that power transmission infrastructure is vitally important to renewable energy substitution and that nuclear and renewable energy can be complementary if the interregional transmission capacity is sufficient in Japan. However, in the United States, Jayadev, Leibowicz [18] found that investing in renewable generation closer to the loads it serves is generally more economical than installing renewables further away to take advantage of higher capacity factors. The studies show that electricity transmission policies under low carbon transitions should be based on local conditions. It is worthwhile to analyze the electricity transmission in China considering the different situations among provinces, such as the economic development level and energy reserves.

Many studies have investigated electricity transmission in China. Zhang, Wang [9] analyzed the application of a cross-region High Voltage Direct Current (HVDC) delivery mode in renewable energy adjustment and noted that interprovincial electricity transmission can effectively reduce renewable energy curtailment. Yang, Zhang [19] pointed that the construction of UHV lines could improve renewable energy substitution with climate policies. Li, Li [20] proposed a new kind of renewable energy capacity adjustment model to study the influence of the utilization level of interprovincial tie lines on the renewable energy capacity and noted that effectively using the interprovincial tie line transmission capacity can significantly enhance the renewable energy accommodation capacity. These studies show that electricity transmission plays a positive role in China's low carbon transition but do not give policy makers a detailed projection of interprovincial electricity patterns in the future.

To analyze future electricity transmission patterns, some studies used energy system model for the power sector. Cheng, Xu [21] developed a ten-region optimization planning model with interregional power transmission for China's power sector and suggested that multi-region optimization better reflects and considers the conditions and challenges in the real world by comparing the results of single-region and multiregion optimizations. Zhang, Ma [22] developed a multi-regional model to optimize long-term electricity generation and transmission in China. Li, Lukszo [23] developed a least-cost multiregion power dispatch model and showed that expanding the interregional transmission grid has a very limited impact on promoting the use of renewable energy substitution by 2030 unless renewable power develops much faster than the current policies require. In addition to energy system model, Zhai, Huang [24] developed a provincial-scale Energy-Carbon Nexus Model in power sector based on input-output analysis (IOA) to quantify the interprovincial carbon and electricity transfers. Tan, Lin [25] employed the mixed complementarity problems (MCP) method and simulating the interactions among participants in the electricity market to investigate the impacts of the construction of the UHV transmission lines. However, these studies separated the power sector or electricity transmission system from the whole energy system, which to some contexts fails to reflect the interactions among different provinces and different sectors.

To obtain a whole energy system perspective, Li and Chen [26] developed a provincial TIMES model, China TIMES-30P, based on the China TIMES (The Integrated MARKAL-EFOM System model) model to analyze changes in China's interprovincial coal transportation and the corresponding emissions under low carbon scenarios. This approach provided a suitable basis for improving the model and analyzing

electricity transmission from a provincial perspective, while the energy system model with five sectors (agriculture, transportation, industry, commercial and residential) provides a whole energy system perspective to analyze electricity transmission under a low carbon transition.

This paper develops a 30-province and multiple-voltage-level electricity transmission model based on the provincial China TIMES model, named China TIMES-30PE (a new version of the China TIMES-30P with multi-voltage level electricity transmission model), to investigate changes in electricity transport patterns from a whole energy system perspective. The impact of a change in the main electricity supply and demand regions on the electricity transport pattern would be better represented by considering the provincial resource endowments and consumption distribution as well as the related transport processes and their costs, which are assessed based on the transport distance, capacity and voltage level. Based on the China TIMES-30PE model, the 30-region energy system which includes the provincial electricity transmission can be optimized to minimize the whole system cost. The modeling period of the China TIMES-30PE is from 2010 to 2050, and the provincial low carbon transition under different scenarios can be simulated during the period. Therefore, future provincial electricity transport would be estimated based on the future provincial power generation mix and electricity demand under the scenarios. Focused on China, this paper provides a method to analyze changes in interprovincial electricity transmission and transmission infrastructure construction in a multiregion energy system that might be applied in other regions where there are sufficient data.

2. Methodology

2.1. The China TIMES-30PE model

The IEA-ETSAP (Energy Technology Systems Analysis Program) developed and maintains the TIMES model, which is a combination of the MARKAL (Market Allocation Model) and the EFOM (Energy Flow Optimization Model). The previous China TIMES-30P model was developed based on the China TIMES [26–32] and China MARKAL [33–35] models. The provincial model contains 30 provincial administrative regions (Taiwan Province, the Tibet Autonomous Region, the Hong Kong Special Administrative Region and the Macao Special Administrative Region are not considered in the model due to statistical limitations). The 30-region energy system model includes energy supply, conversion, transmission and end-use technologies, a modeling period from 2010 to 2050 and a 5-year reporting period. The model can describe the flow of various energy carriers based on the optimal mix of technologies. The historical data used for model calibration were collected from various statistical data sets, reports and official announcements from 2010 to 2015. Based on the socioeconomic development scenarios considered in previous studies [31,32,36], the energy service demand in the model includes five sectors: agriculture, transportation, industry, commercial and residential. With the five sectors and updating the historical data used for model calibration, the China TIMES-30PE model gave a strong base to analyze interprovincial electricity transmission in a whole energy system perspective.

2.2. Interprovincial electricity transmission model

This paper gives a new method to establish an interprovincial multiple-voltage-level electricity transmission model which includes electricity trade matrixes, electricity transmission costs and electricity transmission loss. The electricity transmission model is integrated with the China TIMES-30PE model.

2.2.1. Interprovincial electricity trade matrixes in the electricity transmission model

Based on statistics, reports and official announcements about interprovincial electricity transmission, China's interprovincial electricity

trade matrixes at different voltage levels can be built in the model. Every province is seemed as a single node and can connect with other province by different voltage level electricity transmission processes. Two-direction electricity flows between provinces exist in real life. For instance, in 2011, the electricity flow from Henan to Sichuan reached 5.7 TWh, and the electricity flow from Sichuan to Henan reached 2.6 TWh [8]. The electricity flows in the electricity transmission model built based on the TIMES framework can only be one-way flows because the objective is to minimize the total discounted system cost during the modeling period. Therefore, there are only two processes with different directions in one voltage level between two provinces.

For a peer-to-peer power transmission project between two provinces, it counts in the transmission process between the two provinces in the model. For electricity transmission project among multiple provinces, the model divides a project into multiple subprojects. For example, the Yuheng UHV project, which connects Shaanxi, Shanxi, Hebei and Shandong and transports electricity from Shaanxi and Shanxi to Hebei and Shandong, was abstracted to three subprojects (Shaanxi-Shanxi, Shanxi-Hebei, and Hebei-Shandong) with the same capacities as noted in the statistics. The subproject counts in the transmission process between the two provinces in the model. Therefore, the total capacity of the electricity transmission process in China TIMES-30PE is larger than the actual total capacity.

2.2.2. Interprovincial electricity transmission cost

The interprovincial electricity transmission cost in the China TIMES-30PE model includes the construction cost and operation and maintenance costs. The construction cost of power lines in real life includes lines and transformer substations/converter stations. The line cost associated with the electricity transmission process in China TIMES-30P can be calculated based on the capacity of transmission, the average capacity of lines, the voltage level, the length of the lines and the unit construction prices of power lines. The cost of a transformer substation/converter station in the China TIMES-30PE can be calculated based on the capacity of the electricity transmission process and the unit construction prices of transformer substations/converter stations. The total cost of the electricity transmission process can be calculated based on equation (1). The operation and maintenance cost can be estimated as a fixed proportion of the construction cost.

$$C_{A,B} = l \times \frac{P_{A,B}}{C_{avg}} \times D_{A,B} + r \times P_{A,B} \quad (1)$$

where

$C_{A,B}$: The total cost of the electricity transmission process from province A to province B in China TIMES-30P (10000 yuan);
 l : The unit construction price of power lines (10,000 yuan/km);
 $P_{A,B}$: The capacity of the electricity transmission process from province A to province B (MW);
 C_{avg} : The average capacity associated with a given voltage level for the electricity transmission process (MW);
 $D_{A,B}$: The distance from the capital of province A to the capital of province B (km); and
 r : The unit construction price of a transformer substation/converter station (10,000 yuan/MW).

2.2.3. Interprovincial electricity transmission loss

The transmission loss of interprovincial electricity transmission lines was calculated in the China TIMES-30PE. The typical transmission loss can be calculated using Eq. (2), which ignores complex parameters such as temperature [37]. After removing the irrelevant parameters, the equation can be transformed into Eq. (3). In this case, the transmission loss is proportional to the distance from the capital of province A to the capital of province B.

$$\Delta P_{A,B} = I_{avg}^2 \times \frac{r_0}{n} \times D_{A,B} \times t \quad (2)$$

$$\Delta P_{A,B} = r_1 D_{A,B} \quad (3)$$

where

$\Delta P_{A,B}$: The transmission loss of the line from province A to province B (TWh);
 I_{avg} : The average current on the transmission line (A);
 r_0 : The unit resistance of the transmission line (Ω/km);
 n : The number of splits in the line;
 $D_{A,B}$: The distance from the capital of province A to the capital of province B (km);
 t : The annual operating hours of the transmission line (hours); and
 r_1 : The simplified loss rate of the transmission line.

3. Assumptions and scenarios

3.1. Assumptions and data

3.1.1. Representation of China's power grid in the China TIMES-30PE model

In China's power grid, different voltage levels are used in inter-provincial electricity transmission, including UHV, extra-high voltage (EHV) and high voltage (HV) levels. The voltage level of the UHV lines is above 800 kV, and the voltage level of the EHV lines ranges from 330 kV to 750 kV. The HV lines between provinces only operate at 220 kV. China's mainland power grid is divided into six power grid regions according to the China Electricity Council [8]: the northwest, which includes Xinjiang, Gansu, Qinghai, Ningxia, Tibet, and Shaanxi; the north, which includes Beijing, Tianjin, Hebei, Shanxi, Shandong, and West Inner Mongolia; the northeast, which includes Liaoning, Jilin, Heilongjiang, and East Inner Mongolia; the central, which includes Hubei, Hunan, Jiangxi, Sichuan, Chongqing, and Henan; the east, which includes Shanghai, Jiangsu, Zhejiang, Fujian, and Anhui; and the south, which includes Guangdong, Guangxi, Yunnan, Guizhou, and Hainan, regions. In the China TIMES-30PE, Inner Mongolia is considered a single subregion associated with the north power grid region (Fig. 2).

By 2013, 12 UHV lines and 13 EHV lines were built between power grid regions. There are 58 EHV lines in the northwest power grid region; 27 EHV lines in the north power grid region; 18 EHV lines and 8 HV lines in the northeast power grid region; 1 UHV line and 18 EHV lines in the central power grid region; 4 UHV lines, 17 EHV lines and 2 HV lines in the east power grid region; and 6 UHV lines, 52 EHV lines and 8 HV lines in the south power grid region [38]. EHV lines are the

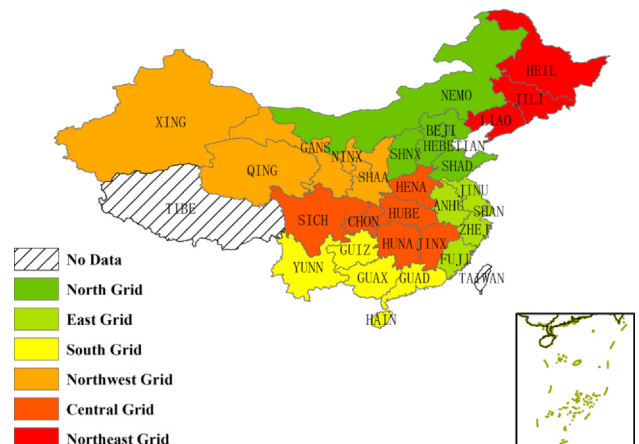


Fig. 2. Power grid region division and provincial abbreviations in China TIMES-30PE.

Table 1
Scenarios in China TIMES-30PE.

Scenarios		Description
Reference	REF	Reference scenario with existing energy policies
Low carbon development scenarios	CP30	NDC scenario 1: Emissions peak at 2030, cumulative emissions in 2010–2050 380Gt
	CPE	NDC scenario 2: Earlier emissions peak, cumulative emissions in 2010–2050 350Gt
	C2D	Well below 2 degrees scenario, cumulative emissions in 2010–2050 290 Gt assumed

main lines used in interprovincial electricity transmission.

UHV, EHV and HV processes were included in the model to describe electricity transport among provinces. Considering the complexity of the grid network, provincial capitals were set as electricity transport centers considering the distance between them and then used to determine the transport distance [26].

The locations, parameters and electricity flows of all UHV and EHV lines built between power grid regions can be found in the announcements of China's National Energy Administration and the annual reports of the China Electricity Council [8,13]. Every province is connected with other provinces in the same power grid region (Fig. 2) by EHV lines and HV lines [8,39]. Because the economically feasible distance for HV line transport is short [39], HV lines can only connect neighboring provinces in the model.

Based on a series of statistical analyses and official reports, the interprovincial electricity flows were calibrated [8,13]. In reports from the National Energy Administration, the interprovincial electricity flows transported by UHV lines were determined [10,12,40], but the available statistics only included the numbers of EHV and HV lines, and data related to the flow or capacity of the lines were unavailable. Therefore, the average capacity and the number of lines were used in the model to calibrate the flows on the EHV and HV lines in the model [8]. The average capacities of the UHV, EHV and HV lines in the China TIMES-30P are 4000 MW, 1500 MW, and 300 MW, respectively [22,39,41,42].

3.1.2. Interprovincial electricity transport cost

The construction cost of power lines in the China TIMES-30PE can be calculated by the method presented in Section 2.2.2. The unit prices of lines and transformer stations at different voltage levels were estimated by the China Electricity Council [13] and are shown in the Appendix. The operation and maintenance costs of UHV, EHV and HV lines are 1.5% [41], 2% [42] and 2% [42] of their construction costs, respectively.

3.1.3. Interprovincial electricity transmission loss rate

The transmission loss rate is proportional to the distance between the related provincial capitals. The transmission loss rate of the UHV line between Shanxi and Henan was 0.7% [38], and was used as the baseline for the UHV lines in the model; that of the EHV line between Shaanxi and Hebei was 2.0%, and was used the baseline for the EHV lines in the model [38], and that of the HV lines in the south was 2.7% and served as the baseline for HV lines in the model [43].

3.2. Scenarios

To explore changes in electricity transport corresponding to low carbon development, the following four scenarios were designed. The economic and social development assumption and energy service demand projection in the scenarios were estimated based on a method that considered balanced development as presented in the literature [44].

The reference scenario is a baseline scenario using the existing energy and economic policies at the national and provincial levels from the 13th Five-year Plan, such as those for industrial structure adjustment, increasing non-fossil energy substitution and reducing carbon

intensity and coal consumption. This scenario addresses the current status of the energy system and the policies needed to meet the goals and commitments for near-term development [31,32,36].

Three low carbon development scenarios were established to study interprovincial electricity transmission under China's NDCs and the 2-degree target in the Paris Agreement. Two low carbon scenarios under the NDCs, CP30 and CPE, were established with the constraint that the national cumulative carbon emissions from 2010 to 2050 are less than 380 Gt and 350 Gt, respectively [1,45]. In these scenarios, the national CO₂ emissions peak in 2030 or in advance. In IPCC AR5 and other studies, a low carbon scenario named C2D was considered to analyze the pathway to achieving the “below 2 degrees” goal in China with the constraint that the national cumulative carbon emissions from 2010 to 2050 are less than 290 Gt [2,46–49] (see Table 1).

4. Results

4.1. Future electricity transmission pattern

In the context of the low carbon transition, the use of fossil fuel will decrease, and renewable energy substitution will accelerate with the strengthening of carbon emission constraints. Carbon emissions will peak in 2040, 2030, 2025 and 2020, and the proportion of non-fossil fuel energy in primary energy use will be 19%, 20%, 21% and 26% in 2030 under the REF, CP30, CPE and C2D scenarios, respectively. The NDC target can be met under the CP30, CPE and C2D scenarios but requires considerable effort. The government should adjust the policies in the 14th Five-year Plan to fully commit to carbon emission reductions. In 2050, electricity generation will reach 11.5 PWh, 11.4 PWh, 11.4 PWh and 12.1 PWh, and the proportion of non-fossil fuel power in electricity generation will be 42%, 52%, 65% and 77% in the REF, CP30, CPE and C2D scenarios, respectively. This finding is consistent with those of other studies. The Energy Research Institute of the National Development and Reform Commission noted that in the high renewable energy penetration scenario, electricity generation would reach 15.2 PWh in 2050 [50]. Cheng developed an optimization model for the power sector and projected that without a carbon emission reduction policy, electricity generation would reach 10.2 PWh in 2050 [21].

In 2015, interprovincial electricity transmission was 948.2 TWh. As shown in Fig. 3, the electricity transmission in 2020 based on REF, CP30, CPE and C2D will be 1523.3 TWh, 1545.2 TWh, 1572.1 TWh and 1646.4 TWh, respectively. Recent statistics indicated that electricity transmission totaled 1295 TWh in 2018 and that the growth rate was approximately 10% from 2007 to 2018; the new electricity transmission model supports these statistics. With the increase in renewable energy power generation, it is necessary to increase construction of transmission infrastructure and transfer cost-effective renewable energy power from western and northern China to various load centers. By 2050, interprovincial electricity transmission will reach 2526.5 TWh, 3299.3 TWh, 3714.2 TWh and 4002.0 TWh under the REF, CP30, CPE and C2D scenarios, respectively. The low carbon scenarios require higher new transmission capacity than the reference scenario, and the transmission demand will increase with the increase in renewable energy power. This result is consistent with the other research, for example, Cheng, Xu [21] built a 10-region power sector optimization planning model to

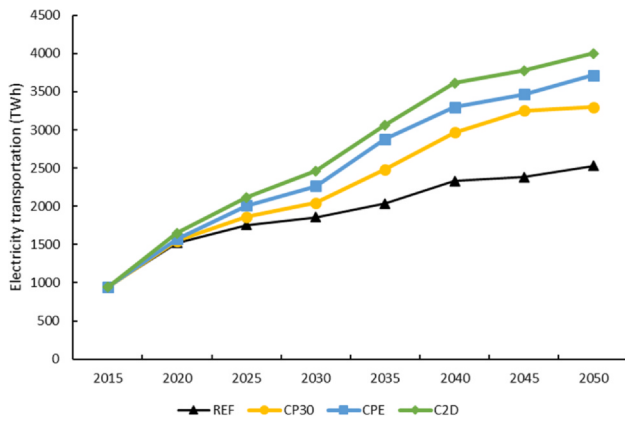


Fig. 3. Amount of interprovincial electricity transmission in 2015–2050 under different scenarios.

access the long-term development of the power sector and projected that the inter-region electricity transmission will reach 3.46PWh in 2050. On the one hand, the results of the China TIMES-30PE suggest that the interprovincial electricity transmission will increase quickly and continuously which is consistent with their results. On the other hand, the results from Cheng and Xu projected greater electricity transmission (among 10 regions). Their model did not include the construction cost of the electricity transmission infrastructure and the China TIMES-30PE considered more detailed data, including construction and operation and maintenance costs, calibrated based on the most recent statistics.

Inter-region (power grid region) electricity transmission capacity will increase in the low carbon transition. In 2030, the capacity of inter-region electricity transmission will reach 317.6 GW, 355.5 GW, 394.6 GW and 445.2 GW under the REF, CP30, CPE and C2D scenarios, respectively (Fig. 4). The electricity transmission capacity increases reported here are in line with the trends reported in other research. For instance, in the study of Yi, Xu [51], the inter-region (12 regions) newly built electricity transmission capacity during 2016–2030 will be 304.2 GW in the renewable energy scenario (renewable energy share of 20% in primary energy). In the China TIMES-30PE, the newly built inter-region electricity transmission capacity during 2016–2030 will be 244.3 GW and 294.9 GW in the CP30 and CPE scenarios, respectively. On the one hand, the results of the China TIMES-30PE suggest that the capacity of interprovincial electricity transmission will increase quickly and continuously which is consistent with the results of Yi, Xu [51]. On the other hand, different regional settings mean that the results from the China TIMES-30PE are smaller than their results in the similar

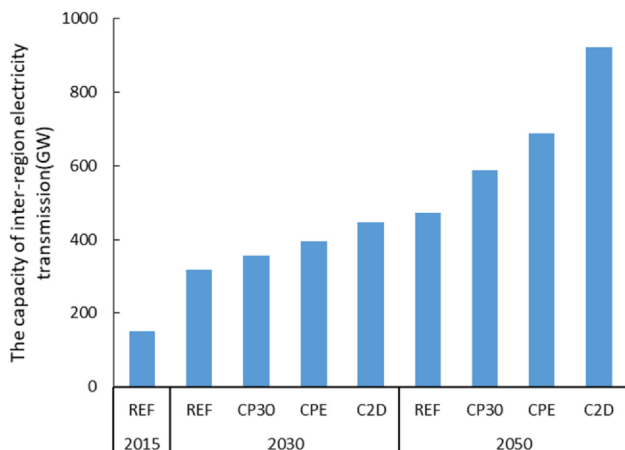


Fig. 4. The capacity of inter-region electricity transmission under different scenarios (GW).

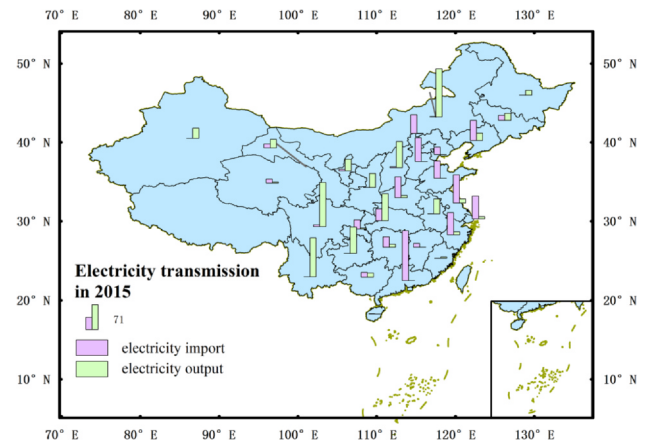


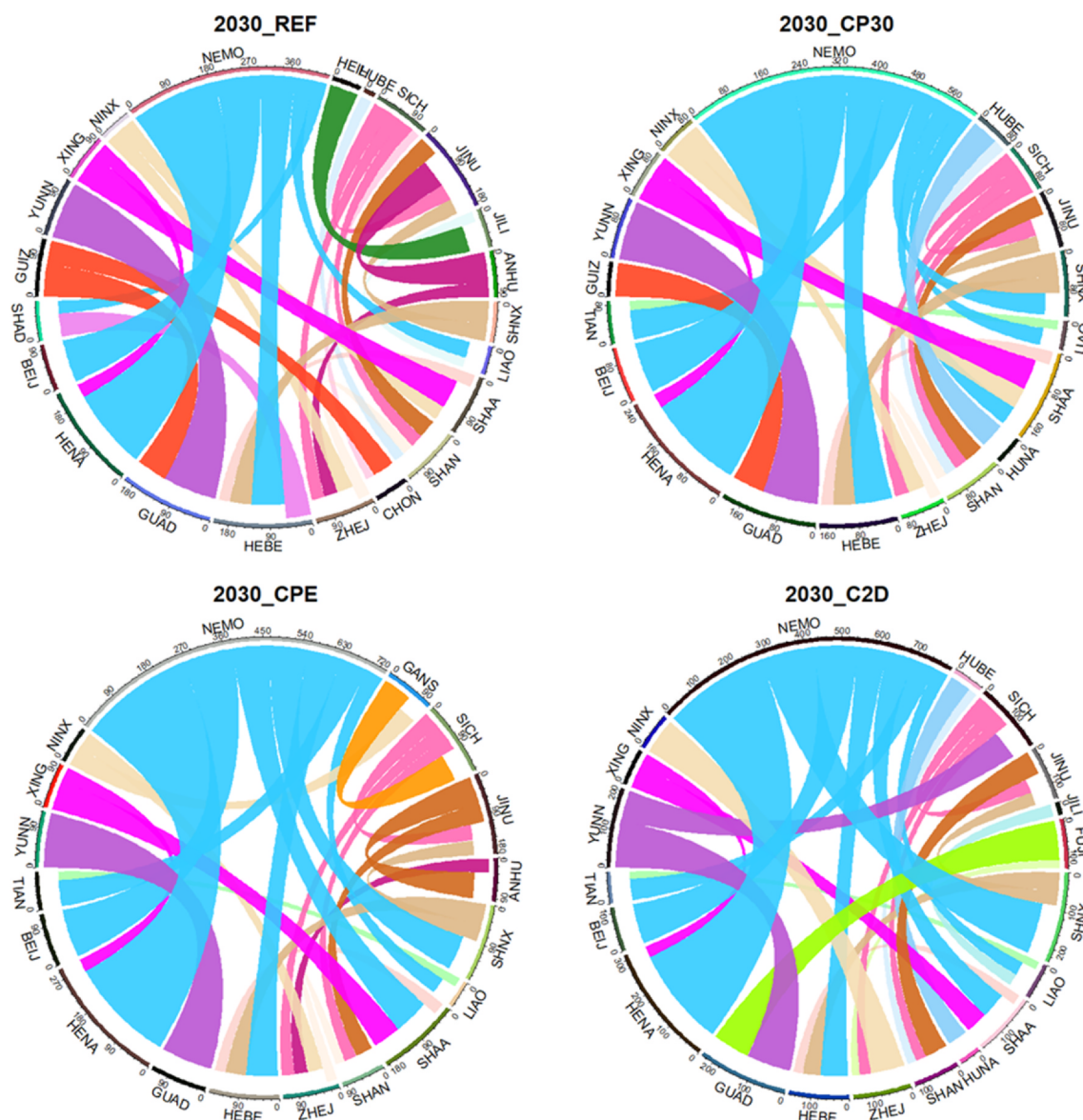
Fig. 5. Interprovincial electricity transmission in 2015 (TWh).

scenarios (CP30, CPE and their renewable scenarios).

In 2050, the capacity of inter-region electricity transmission will be 470.9 GW, 588.9 GW, 687.3 GW and 922.3 GW under the REF, CP30, CPE and C2D scenarios, respectively (Fig. 4). These results are consistent with those from other conducted research. A study from Energy Research Institute of the National Development and Reform Commission showed that the newly built inter-region electricity transmission capacity will be 575.0 GW during 2016–2050 under the high renewable energy penetration scenario (carbon emissions peak in 2025). In the China TIMES-30PE, the newly built inter-region electricity transmission capacity will be 537.0 GW in the CPE scenario for 2016–2050 which is consistent with the research.

From the provincial perspective, as shown in Fig. 5, the provinces that exported more than 100 TWh of electricity were Inner Mongolia, Sichuan, and Yunnan, accounting for 14.6%, 13.4%, and 11.7%, respectively, of interprovincial electricity exports nationwide. The provinces with a high capacity in coal power or hydropower were the main electricity exporters rather than the provinces with high wind and solar reserves.

In the future, the northern and northwestern provinces will be the major providers of electricity, and there will be a continuous increase in electricity transportation after 2015 under the reference scenario and low carbon emission scenarios. The largest electricity output province, Inner Mongolia, is a coal energy base that has high renewable energy reserves as well. In 2015, coal electricity generation in Inner Mongolia reached 340.9 TWh, accounting for 88.4% of provincial electricity generation. Inner Mongolia is near load centers, such as the Beijing-Tianjin-Hebei region (Beijing, Tianjin and Hebei) and Shandong. In the 13th Five-year Plan, Inner Mongolia was projected to be a major energy export base. By 2019, 7 UHV lines has been built to transport electricity to the Beijing-Tianjin-Hebei region, Shandong, Jiangsu and Shanxi. The electricity exported from Inner Mongolia could reach 498.1 TWh, 667.9 TWh, 795.2 TWh and 828.3 TWh under the REF, CP30, CPE and C2D scenarios, respectively, by 2030 (Fig. 6) and 585.9 TWh, 1159.4 TWh, 1061.5 TWh and 897.1 TWh by 2050 (Fig. 7). Under the strict carbon emission constraint in C2D, the decrease in coal power generation will be accompanied by a decrease in electricity transport. However, the electricity transported from Xinjiang, which has large reserves of coal, solar and wind, will total 222.2 TWh, 311.8 TWh and 335.5 TWh under the CP30, CPE and C2D scenarios, respectively, in 2050. In 2018, electricity transmission from Xinjiang to Shaanxi reached 16.3 TWh, and electricity transmission from Shaanxi to Hebei reached 22.4 TWh. Therefore, Shaanxi can be considered as a transfer station for electricity transmission from Xinjiang to the Beijing-Tianjin-Hebei load center. Additionally, Shaanxi was a coal power base, but electricity exports in 2018 were 44% lower than in 2015. Under the low carbon transition, there will continue to be a decrease in electricity exports from Shaanxi.



The electricity exports from Ningxia will be influenced by carbon emission constraints, as in Inner Mongolia. The electricity transport in Gansu and Qinghai, which account for large proportions of renewable energy generation, will increase with reductions in carbon emissions.

In Northeast China, Heilongjiang and Liaoning will transport electricity to the Beijing-Tianjin-Hebei region under the low carbon scenarios. In 2030 and 2050, Heilongjiang will transport electricity to Jilin under the reference scenario and to Hebei under the low carbon scenarios. The share of renewable energy in total electricity generation in Heilongjiang will increase from 17% in the reference scenario to 30%, 50% and 61% under CP30, CPE and C2D in 2050, respectively. Additionally, in 2050, the electricity transported from Heilongjiang will reach 117.0 TWh, 165.7 TWh, 229.0 TWh and 316.4 TWh for the REF, CP30, CPE and C2D scenarios, respectively. With the development of nuclear power and renewable power, Liaoning will transport electricity to Tianjin and Hebei in 2030 and 2050.

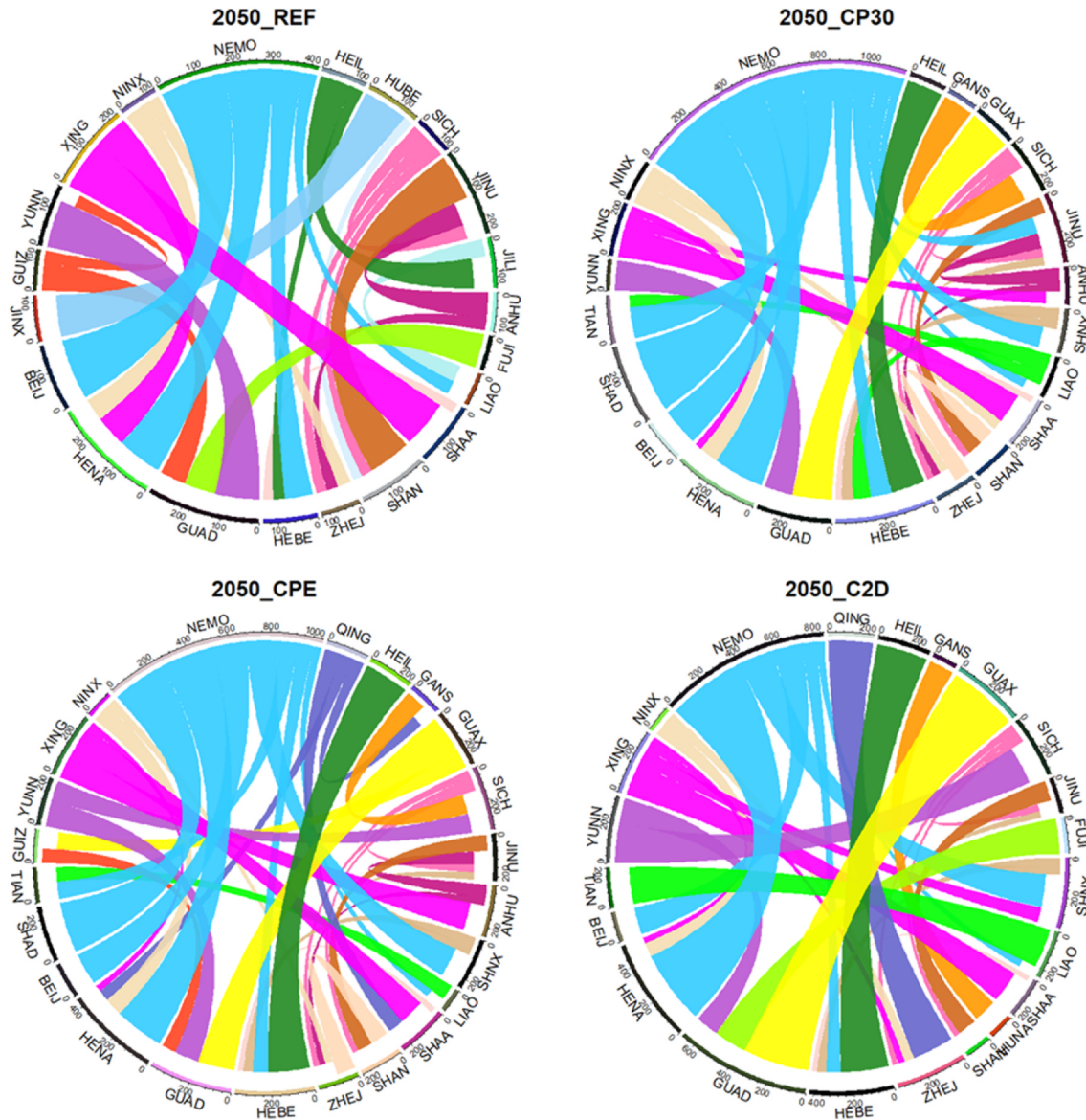


Fig. 7. Change in the electricity transportation pattern in 2050 under different scenarios (TWh).

Yunnan and Guizhou are the main electricity suppliers for Guangdong Province, accounting for more than 38.8% and 22.4%, respectively, under the low carbon scenarios. With the reduction in power production from fossil energy, mainly due to nuclear and hydropower, Guangxi will replace Guizhou and become the largest electricity provider for Guangdong in 2050 and electricity transmission will reach 175.8 TWh, 189.9 TWh and 363.9 TWh under the CP30, CPE and C2D scenarios, respectively. Fujian will provide electricity to Guangdong based on coal power in the reference scenario and nuclear power in the low carbon scenarios.

4.2. Development of UHV lines in 2030 and 2050

With the increasing proportion of renewable electricity, the mismatch of power generation from renewable energy and electricity demand distribution will increase the long-distance electricity transmission demand.

On the one hand, wind and solar power generation only accounted for 3.9% of national electricity generation in 2015 (Fig. 8). On the other hand, inter-regional connections were just completed for the Chinese grid, and there was insufficient interprovincial electricity transmission

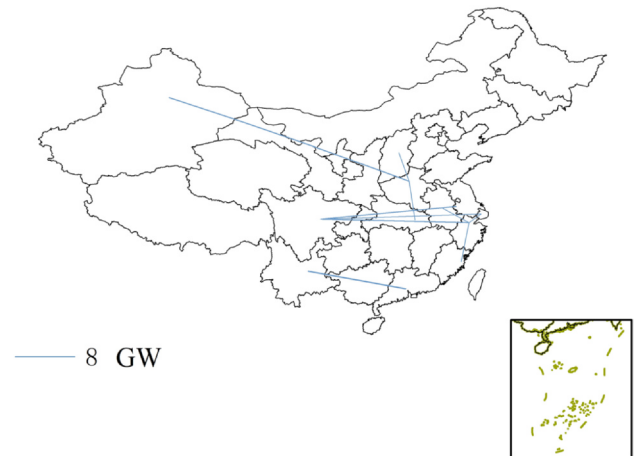


Fig. 8. Transmission capacity of operating UHV lines in 2015.

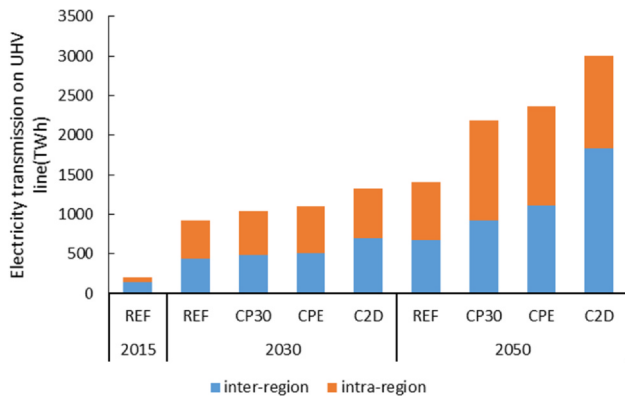


Fig. 9. Interregional and intraregional electricity transmission on UHV lines.

capacity to export electricity from the provinces that faced serious wind and solar power curtailment issues, such as Gansu. To meet the carbon emission reduction target of the NDCs or the Paris Agreement, it is necessary to develop renewable energy sources and improve inter-provincial electricity transmission capacity.

With the advantages of long-distance transport, high capacity and a low line loss, there will be a continuous increase in electricity transmission via UHV lines after 2015 (Fig. 9). The electricity transmission by UHV lines is projected to reach 924.4 TWh, 1037.1 TWh, 1098.7 TWh and 1321.3 TWh in 2030 and 1410.5 TWh, 2186.7 TWh, 2365.5 TWh and 3004.7 TWh in 2050 under the REF, CP30, CPE and C2D scenarios, respectively. Interregional long-distance UHV electricity transmission will potentially reach 925.5 TWh, 1104.9 TWh and 1830.6 TWh under the CP30, CPE and C2D scenarios, respectively, which will be larger than the 674.7 TWh under the reference scenario.

In 2018, the operating capacity of UHV lines was 146.9 GW, and the electricity transmission on UHV lines was 398 TWh. The load rate in 2018 was used to calibrate the electricity transmission capacity demand for UHV lines in the model. As shown in Fig. 10, there will be a continuous increase in the electricity transmission capacity demand of UHV lines under the reference and low carbon emission scenarios. In 2030, the electricity transmission capacity demand of UHV lines is projected to reach 355.0 GW, 390.3 GW, 418.0 GW and 515.4 GW in the REF, CP30, CPE and C2D scenarios, respectively, with values of 595.6 GW, 843.3 GW, 917.1 GW and 1198.0 GW in 2050 (see Fig. 10).

The UHV lines in Inner Mongolia were built to transport electricity to Henan, Tianjin, Shandong, Jiangsu, Shanxi and Beijing. The electricity transmission demand for UHV lines from Inner Mongolia to other areas will potentially reach 133.1 GW, 175.9 GW, 211.6 GW and 245.8 GW in 2030 (Fig. 11) and 207.0 GW, 390.1 GW, 360.4 GW and 316.8 GW in 2050 (Fig. 12) under the REF, CP30, CPE and C2D scenarios, respectively. Inner Mongolia-Henan, Inner Mongolia-Shandong, Inner Mongolia-Beijing and Inner Mongolia-Tianjin will be the major

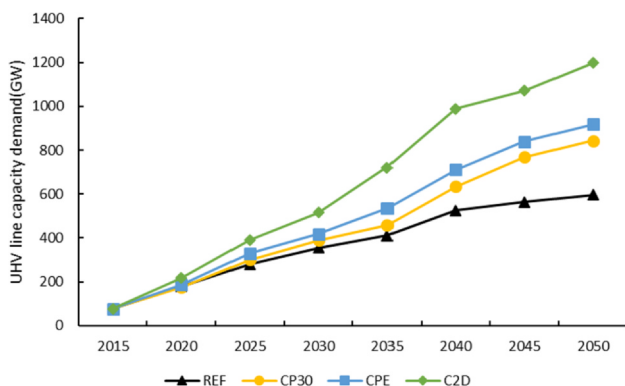


Fig. 10. UHV line capacity demand under different scenarios.

electricity transmission paths from Inner Mongolia. The UHV lines in Xinjiang were built to transport electricity to Henan, Anhui, Shaanxi and Zhejiang. Xinjiang-Shaanxi, Xinjiang-Zhejiang and Xinjiang-Henan will be the major electricity transmission paths from Xinjiang. The UHV electricity transmission capacity demand of Shaanxi will increase continuously through 2030 and through 2050. The electricity transmission capacity demand of Xinjiang-Zhejiang, the longest electricity transmission path, could reach 13.3 GW in 2050 under the C2D scenario, and electricity transport on the path will potentially total 37.6 TWh.

The UHV lines in Northeast China will mainly form an electricity transmission path from the northeast region to the Beijing-Tianjin-Hebei region under the low carbon emission scenarios. Electricity transmission on the UHV lines will contribute more than 90% of the total electricity transported from the northeast region to the Beijing-Tianjin-Hebei region under low carbon scenarios in 2050. Heilongjiang-Hebei and Liaoning-Tianjin will be the major electricity transmission paths in Northeast China, and the capacity demands of Heilongjiang-Hebei and Liaoning-Tianjin will potentially reach 95.2 GW and 66.7 GW, respectively, under the C2D scenario in 2050 (Fig. 12).

The Yangtze River Delta, which includes Anhui, Shanghai, Jiangsu and Zhejiang, will be the major destination of UHV lines in the central and eastern provinces. Inner Mongolia, Xinjiang, Ningxia, Qinghai, Gansu and Sichuan will transport electricity to the Yangtze River Delta. The capacity demand of the UHV lines to the Yangtze River Delta will be 103.5 GW, 144.7 GW, 161.3 GW and 96.9 GW under the REF, CP30, CPE and C2D scenarios, respectively, in 2050.

In South China, Guangdong, the load center, will be the major destination of UHV lines. Yunnan-Guangdong and Guangxi-Guangdong will be the major electricity transmission paths of Guangdong, and the demand of the UHV lines to Guangdong will be 60.0 GW, 87.5 GW, 101.6 GW and 238.3 GW under the REF, CP30, CPE and C2D scenarios, respectively, in 2050.

5. Conclusion and discussion

To analyze the changes in China's future electricity transmission and the development of ultrahigh-voltage lines, this paper improved a bottom-up energy system model covering 30 provincial administrative regions in China to incorporate a provincial electricity transmission module. A method to model the inter-region electricity transmission in a multiregion energy system was established that might be applied to other regions if there are sufficient data.

The low carbon transition will accelerate new and renewable energy development in China. Total electricity generation will reach 11.5 PWh, 11.4 PWh, 11.4 PWh and 12.1 PWh and the proportion of non-fossil fuel power production will reach 42%, 52%, 65% and 77% under the REF, CP30, CPE and C2D scenarios, respectively, in 2050. The mismatch between the electricity demand and renewable energy reserves will result to mismatch between electricity demand and power generation in the low carbon transition, leading to increased electricity transmission. The maturation of ultrahigh-voltage technology provides a technical basis for long-distance and large-capacity electricity transmission. The ultrahigh-voltage lines will play an important role in electricity transmission during the low carbon transition. In 2050, interprovincial electricity transmission is expected to increase to 2526.5 TWh, 3299.3 TWh, 3714.2 TWh and 4002.0 TWh under the REF, CP30, CPE and C2D scenarios, respectively, and UHV electricity transmission is expected to reach 1410.5 TWh, 2186.7 TWh, 2365.5 TWh and 3004.7 TWh, respectively. In addition, the capacity of inter-region (power grid region) electricity transmission will increase in the low carbon transition.

With the increase in electricity transmission and the development of ultrahigh-voltage lines, the overall pattern of electricity transmission will likely become increasingly complex in the future. Inner Mongolia will change from mainly serving Liaoning and Hebei to serving North China, especially the Beijing-Tianjin-Hebei region, Central China and East China. Northwest China will mainly providing electricity to Central

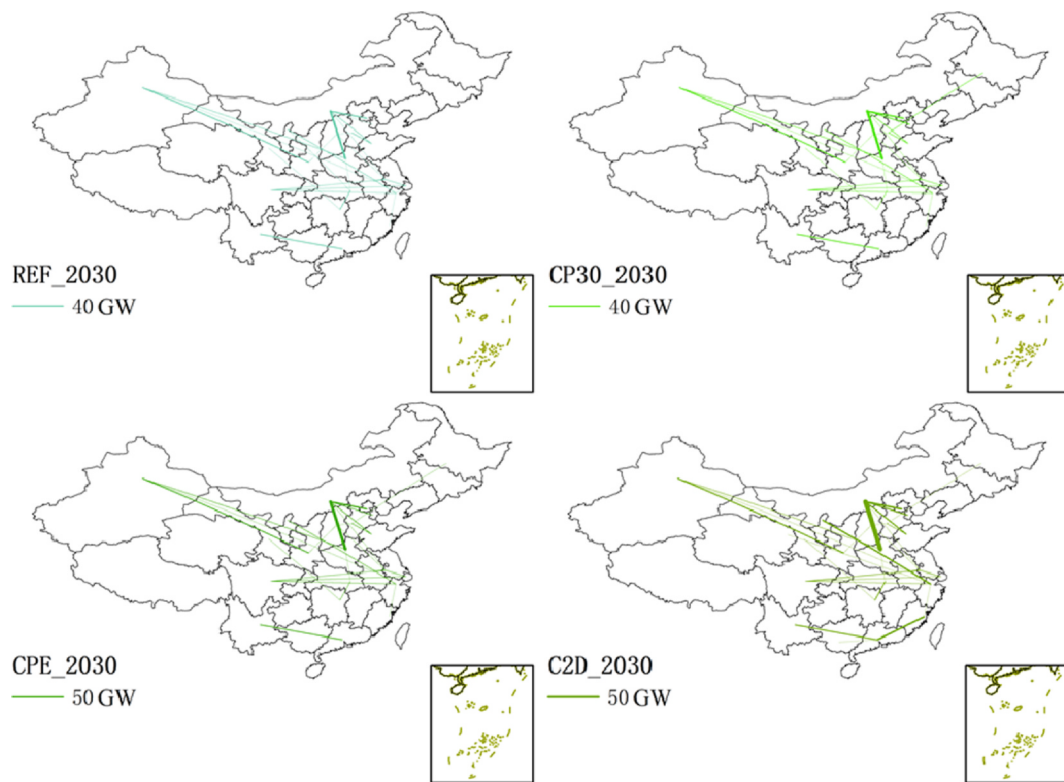


Fig. 11. Interprovincial electricity transmission capacity demand for UHV lines in 2030.

China and East China instead of Northwest and Central China. The northeastern provinces will provide electricity to the Beijing-Tianjin-Hebei region in the future. The southwestern provinces, with abundant hydropower, will continue to provide electricity top Central China and

East China, while Guangdong will continue to import electricity from Yunnan, Guizhou, Guangxi and Fujian.

Low carbon transition requires substantial decrease of power generation from coal power-based provinces. There will still be an increase

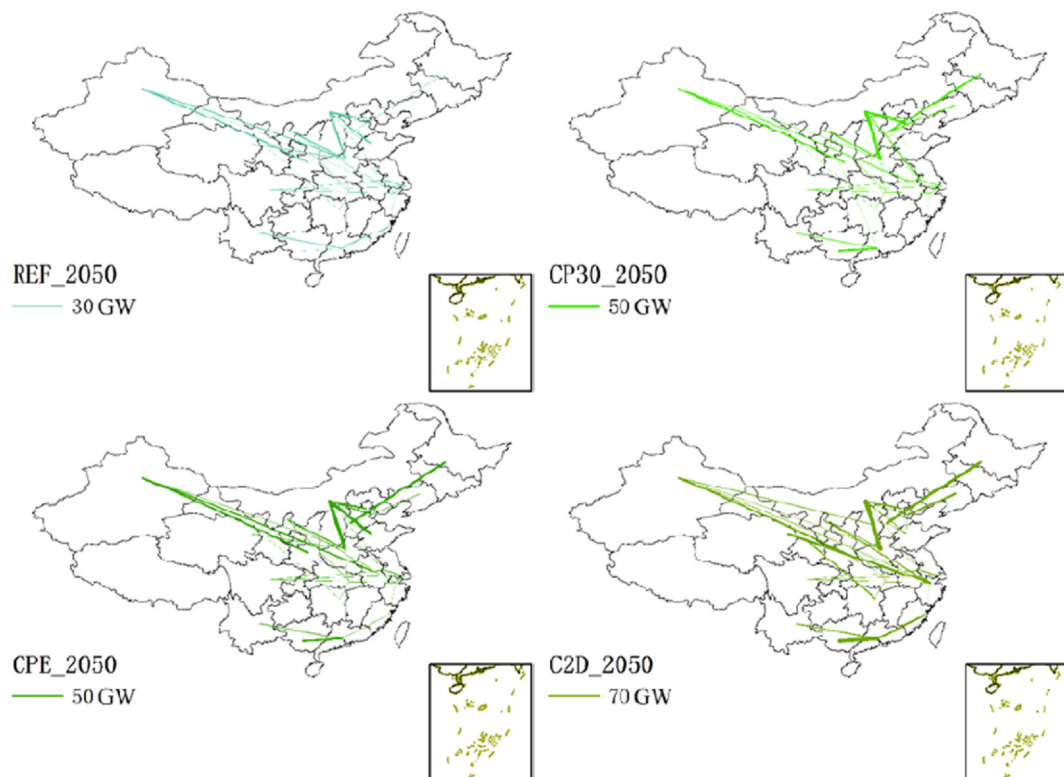


Fig. 12. Interprovincial electricity transmission capacity demand for UHV lines in 2050.

Table A1
Provinces and abbreviations.

Province	Abbreviation	Province	Abbreviation
Beijing	BELJ	Hunan	HUNA
Tianjin	TIAN	Guangdong	GUAD
Hebei	HEBE	Guanxi	GUAX
Shanxi	SHNX	Hainan	HAIN
Inner Mongolia	NEMO	Chongqing	CHON
Liaoning	LIAO	Sichuan	SICH
Jilin	JILI	Guizhou	GUIZ
Heilongjiang	HEIL	Yunnan	YUNN
Shanghai	SHAN	Shaanxi	SHAA
Jiangsu	JINU	Gansu	GANS
Zhejiang	ZHEJ	Qinghai	QING
Anhui	ANHU	Ningxia	NINX
Fujian	FUJI	Xijiang	XING
Jiangxi	JINX	Xizang	TIBE
Shandong	SHAD	Hongkong	HK
Henan	HENA	Macao	Macao
Hubei	HUBE	Taiwan	TAIWAN

Table B1
Data of power lines on different voltage levels in China TIMES-30PE.

Process	Average capacity (MW)	Unit price of Lines (10,000 yuan/ km)	Unit price of transformer/ converter station (10,000 yuan/ MW)	Operation and maintenance cost rate
UHV	4000	468.35	579.95	1.5%
EHV	1500	248.80	117.04	2%
HV	300	111.49	224.24	2%

in electricity transmission in provinces with considerable non-fossil fuel energy substitution, such as Inner Mongolia and Xinjiang. However, in provinces with limited non-fossil fuel energy substitution, such as Shaanxi, the electricity transmission pattern will change from exports to imports during the low carbon transition. With economic development, the electricity exported from provinces such as Sichuan will decrease, while the gap between electricity demand and generation will increase in provinces with insufficient non-fossil fuel energy generation.

The impacts of the low carbon transition will increase not only interprovincial electricity transmission but also the transmission distance. Under these conditions, the demand for ultrahigh-voltage electricity transmission capacity will grow as renewable energy substitution increases and potentially reach 595.6 GW, 843.3 GW, 917.1 GW and 1198.0 GW in 2050. The ultrahigh-voltage lines will form large-capacity and long-distance electricity transmission paths that connect electricity generation bases and load centers, such as the North-East and Northwest-Central pathways.

A new method of analyzing interprovincial electricity transmission in a whole energy system perspective by building an interprovincial electricity transmission module integrated with an improved 30-province energy system model (China TIMES-30PE) is developed. Electricity demands and resource endowments vary across provinces, and the model employed here can effectively represent these provincial differences and interactions, thereby providing a detailed characterization of future electricity transmission patterns considering the conditions of electricity transmission and infrastructure construction.

The vigorously developing global renewable energy under global low carbon transition are paramount driving forces for building the Global Energy Interconnection (GEI). The electricity transmission will

be the most important bridge in GEI. Based on integrated assessment models, the global electricity transmission and infrastructure construction could be further explored from a system perspective in future studies according to the method in this paper.

During the low carbon transition, especially in the 2-degree scenario, the increased integration of wind and photovoltaic power into grid might lead to large uncertainty caused by variable solar irradiation and wind speeds. The potential impacts on electricity transmission and transmission infrastructure construction should be further explored in future studies.

CRediT authorship contribution statement

Qiang Zhang: Methodology, Data curation, Visualization. **Wenyang Chen:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

See [Tables A1 and B1](#).

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